

SYSTEM DESIGN

Analytics and planning on Fiori

System design, and how it compares with SAP Analytics Cloud

One system for plan and data

1

The need

Planning and analysis on S/4HANA data today means a second product with its own license, connections and security.

2

The question

How much of the SAC working model can run inside the ERP landscape, under the ERP's own authorizations?

3

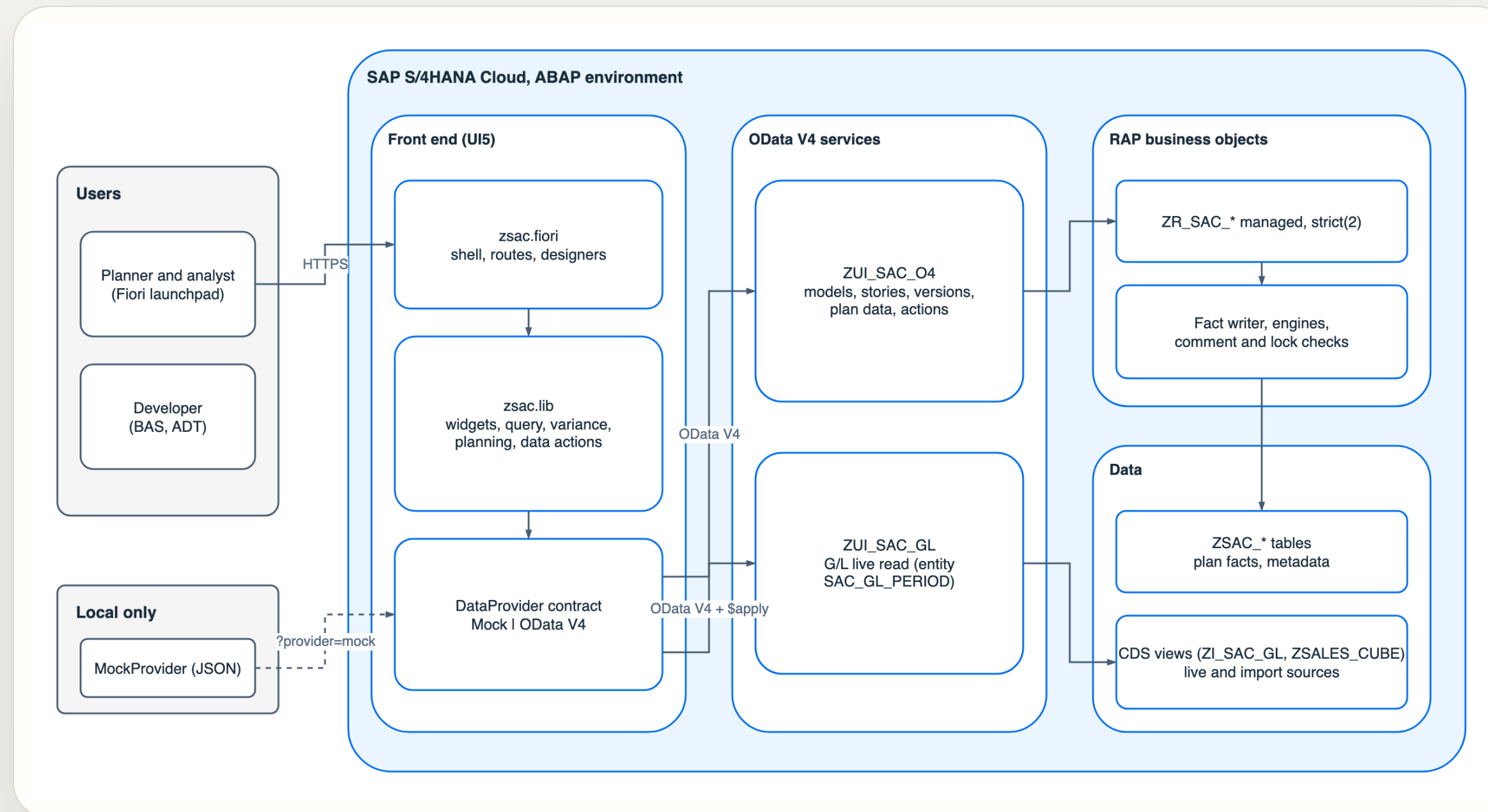
The answer so far

Stories, models, planning with versions, data actions, multi actions, CDS sources and variance analysis, tested on a real system.

What it does

AREA	CAPABILITY
Stories	Pages, designer, charts, KPIs, tables, filters
Models	Measures, dimensions, hierarchies, CSV and CDS sources
Planning	Editable table, spreading, versions, publish, comments
Data actions	Copy, allocate, scale, delete; validate, trace, run
Multi actions	Ten step types: lock, forecast, import, API
CDS sources	Any CDS view as a live or an import source
Variance	Why a measure changed, by every dimension

The big picture



Three design principles

Modular

The library `zsac.lib` serves any Fiori app. Widgets and data sources are plug-ins registered in one place.

Metadata is data

Models, stories, actions and files are rows, edited by the same screens that show them.

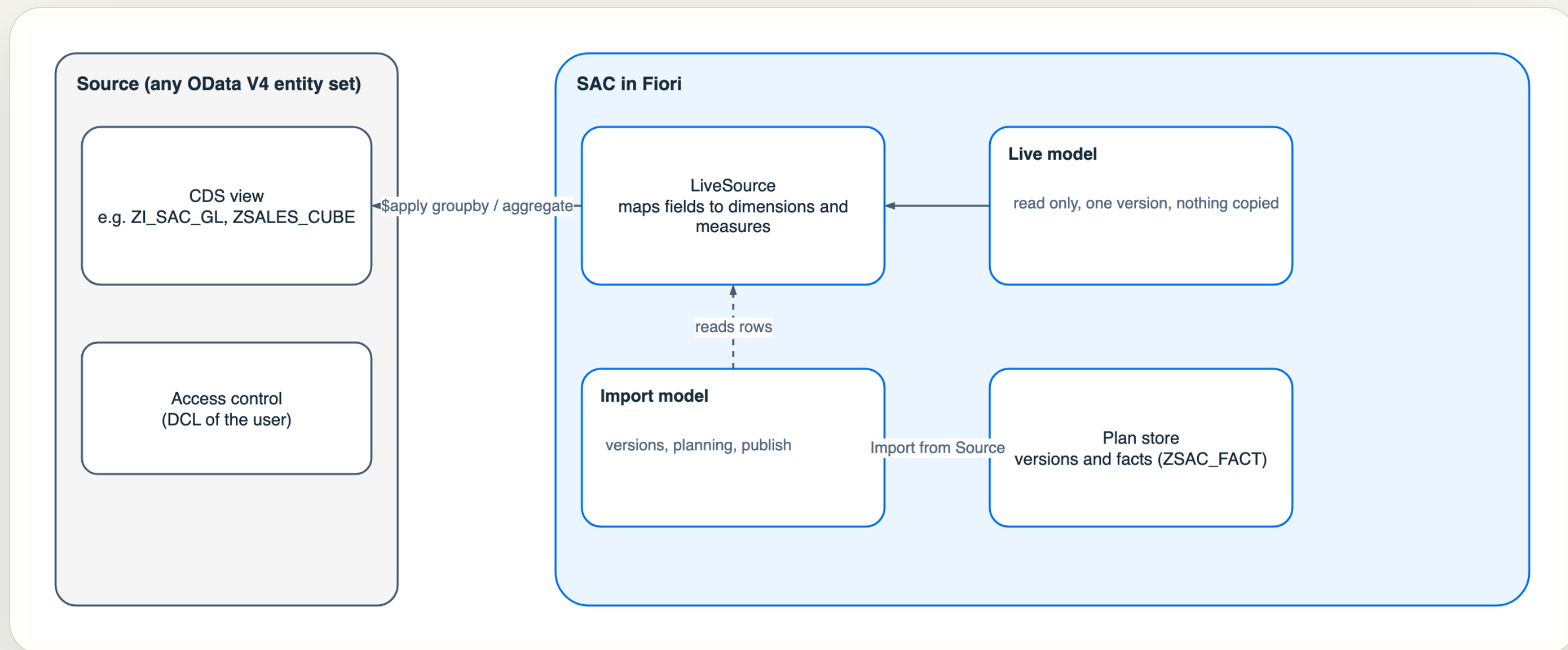
One provider contract

Mock for development and tests, OData V4 for the backend, CDS views for outside data.

DATA

Live and import sources

The service aggregates with \$apply; if it cannot, the app reads the rows and aggregates itself.



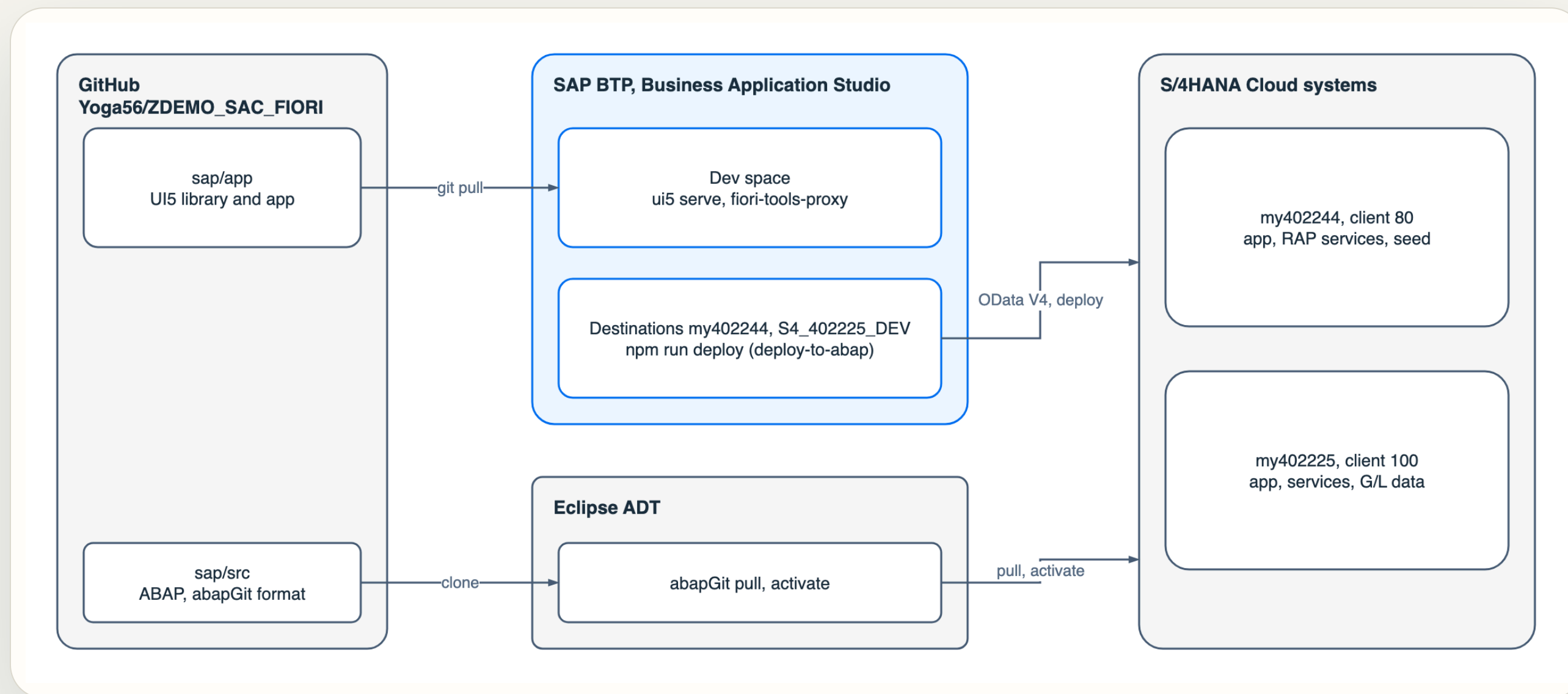
ABAP RAP at a glance

PART	DESIGN
Tables	Fifteen ZSAC_* tables: files, stories, models, versions, facts, actions, runs
Business objects	Managed, strict(2); one request writes a root and its children
Actions	Version: create private, publish, revert. Facts: bulk write and delete
Server rules	Locked versions refuse writes; model structure is checked on save
Generated	One spec produces CDS, behavior, projections and abapGit files
Service	ZUI_SAC_O4 for the app, ZUI_SAC_GL for the G/L view

DELIVERY

Landscape and delivery

The same code runs on mock data, in Business Application Studio and deployed to the system.



Security and operations

In place

Calls run under the user's own session; CDS access control applies to every read.

In place

Locked versions refuse writes on the server and in every engine.

In place

Every action run is recorded: status, counts, duration, user.

Open

No owner or sharing rules for stories and models yet.

Open

No row level security on plan data; actions run in the browser.

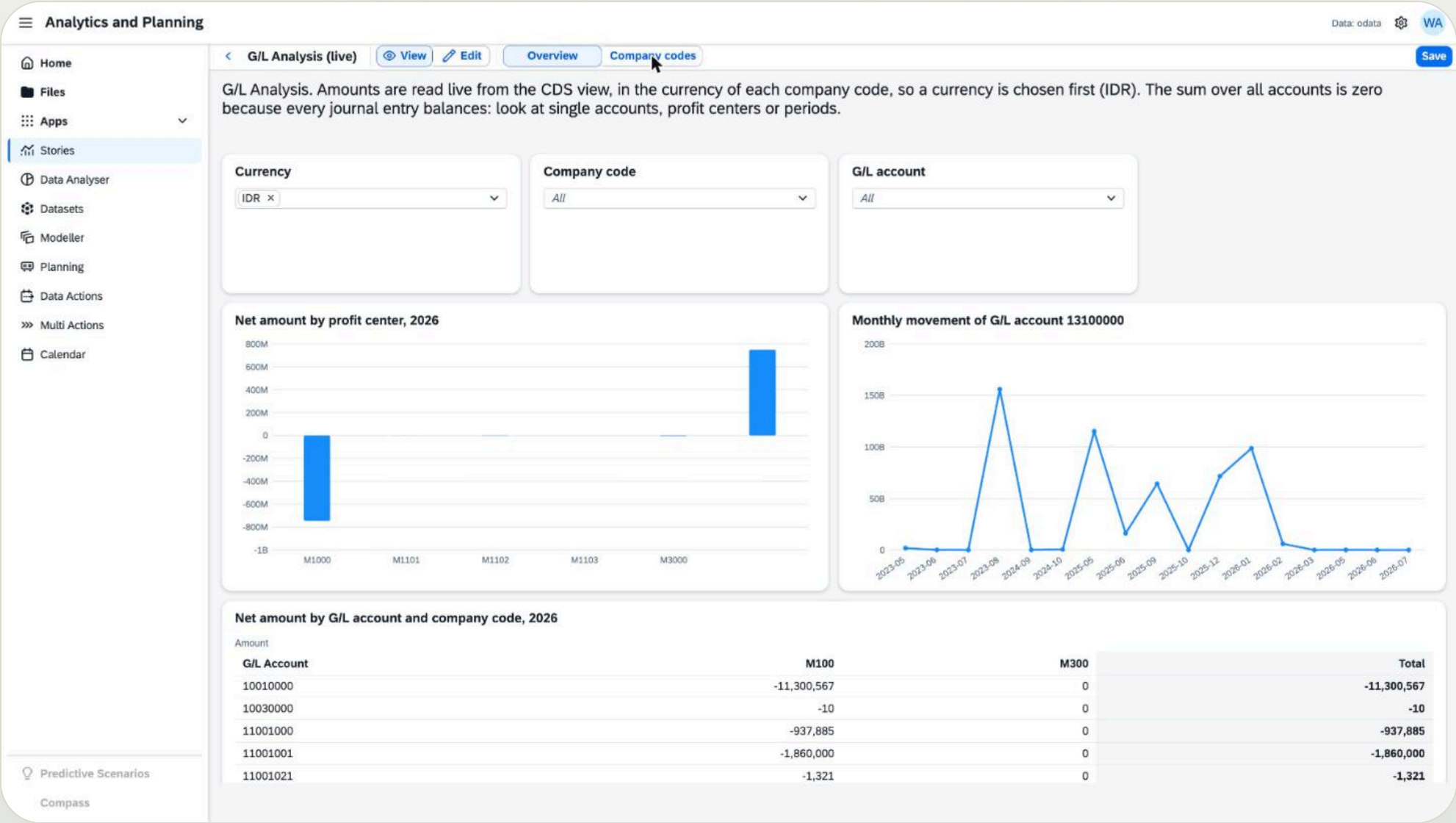
Open

Content moves only through abapGit and the seed class.

REAL SYSTEM · G/L

Live G/L analysis

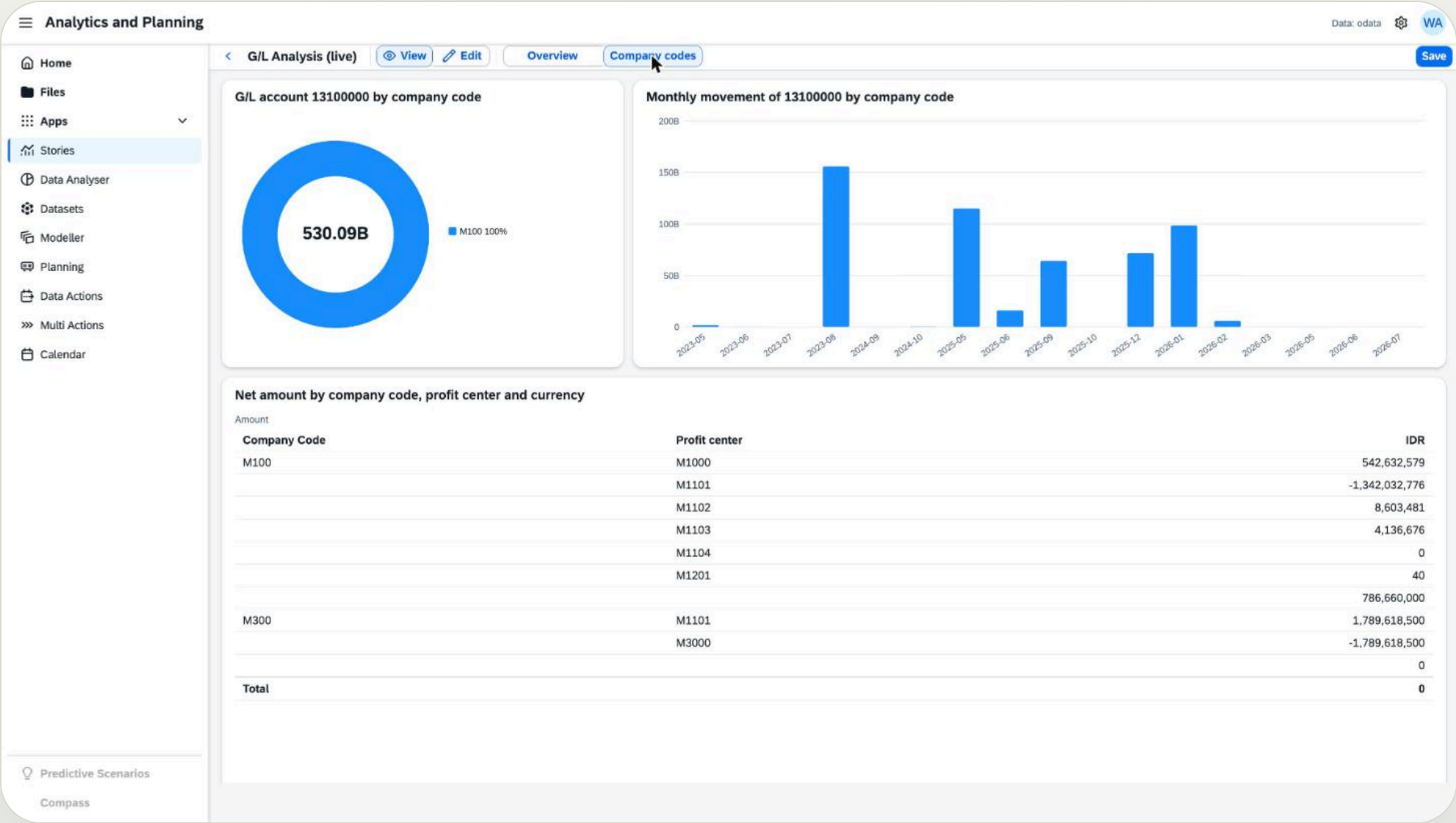
A story on the G/L CDS view with 1,092 rows, read live from the S/4HANA Cloud system.



Story G/L Analysis, page 1, on system my402225

Company codes

The second page breaks the same live data down by company code.

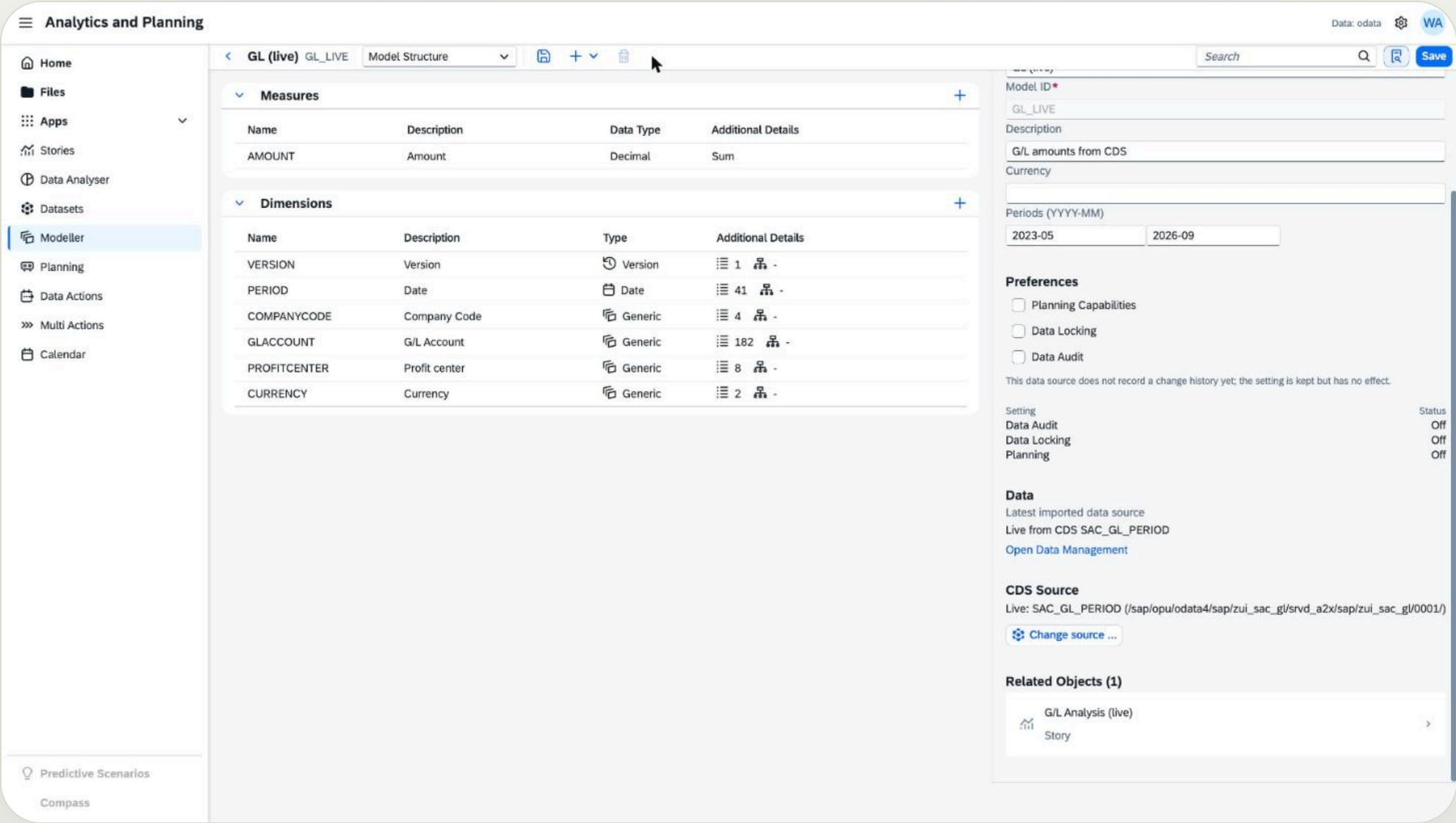


Story G/L Analysis, page 2

REAL SYSTEM · G/L

The model on a CDS view

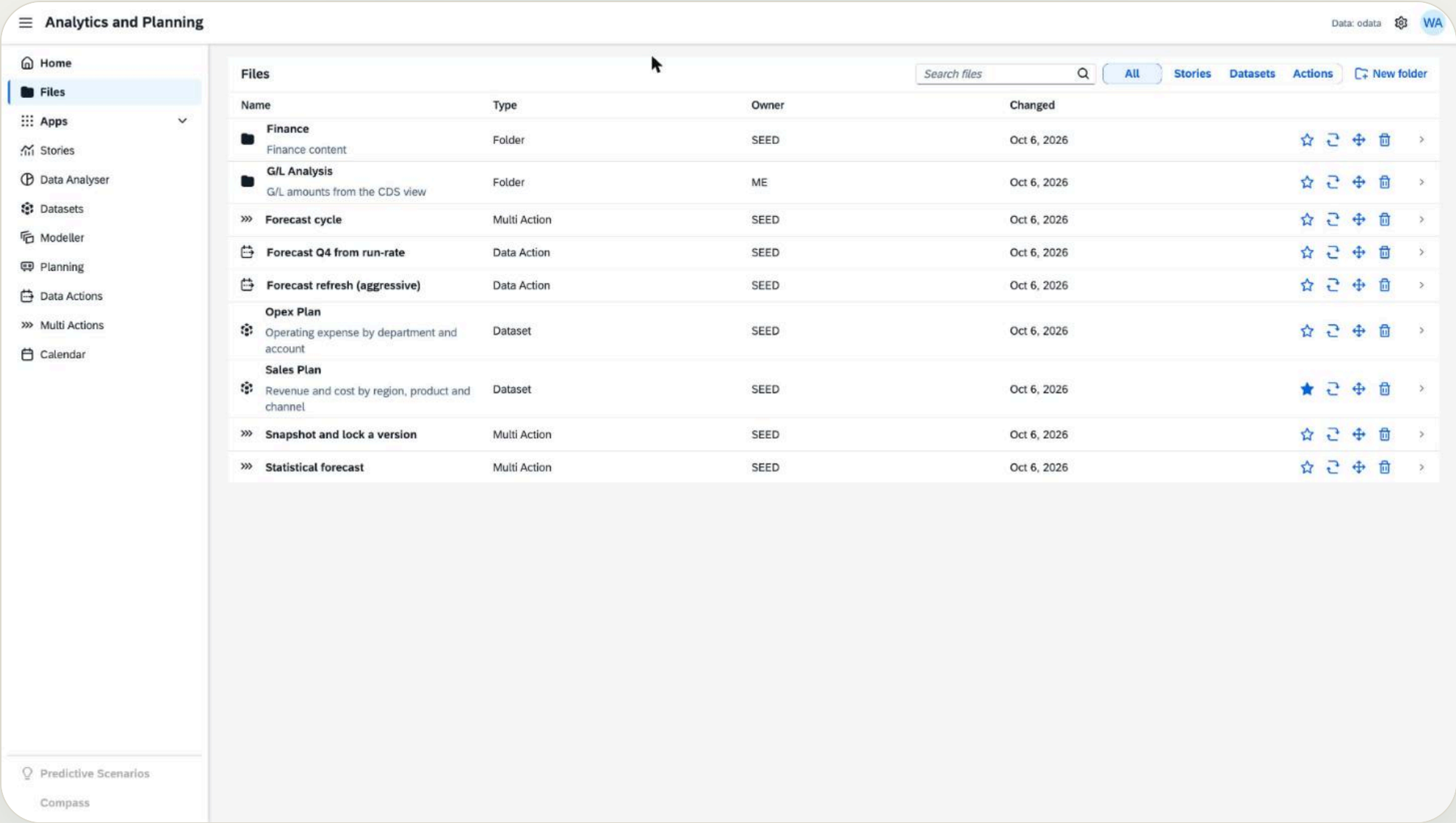
The live model maps fields of the service to dimensions and measures. The source panel shows the service, entity and mapping.



Model GL_LIVE in the Modeller

Files and folders

Models and stories are files. The G/L model, plan model and story sit in their own folder.

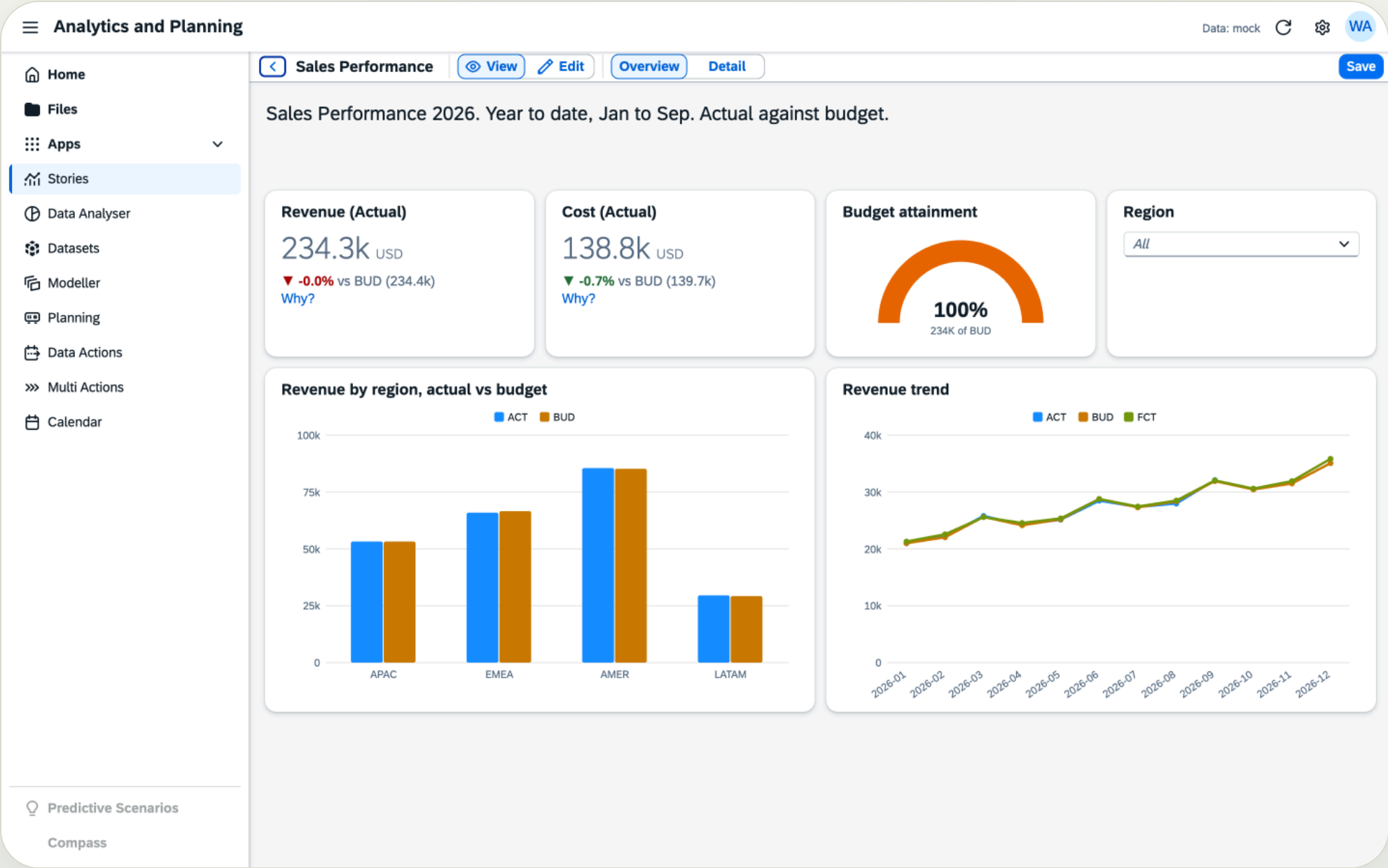


Files: Finance and G/L Analysis folders

SAMPLE DATA

Stories and charts

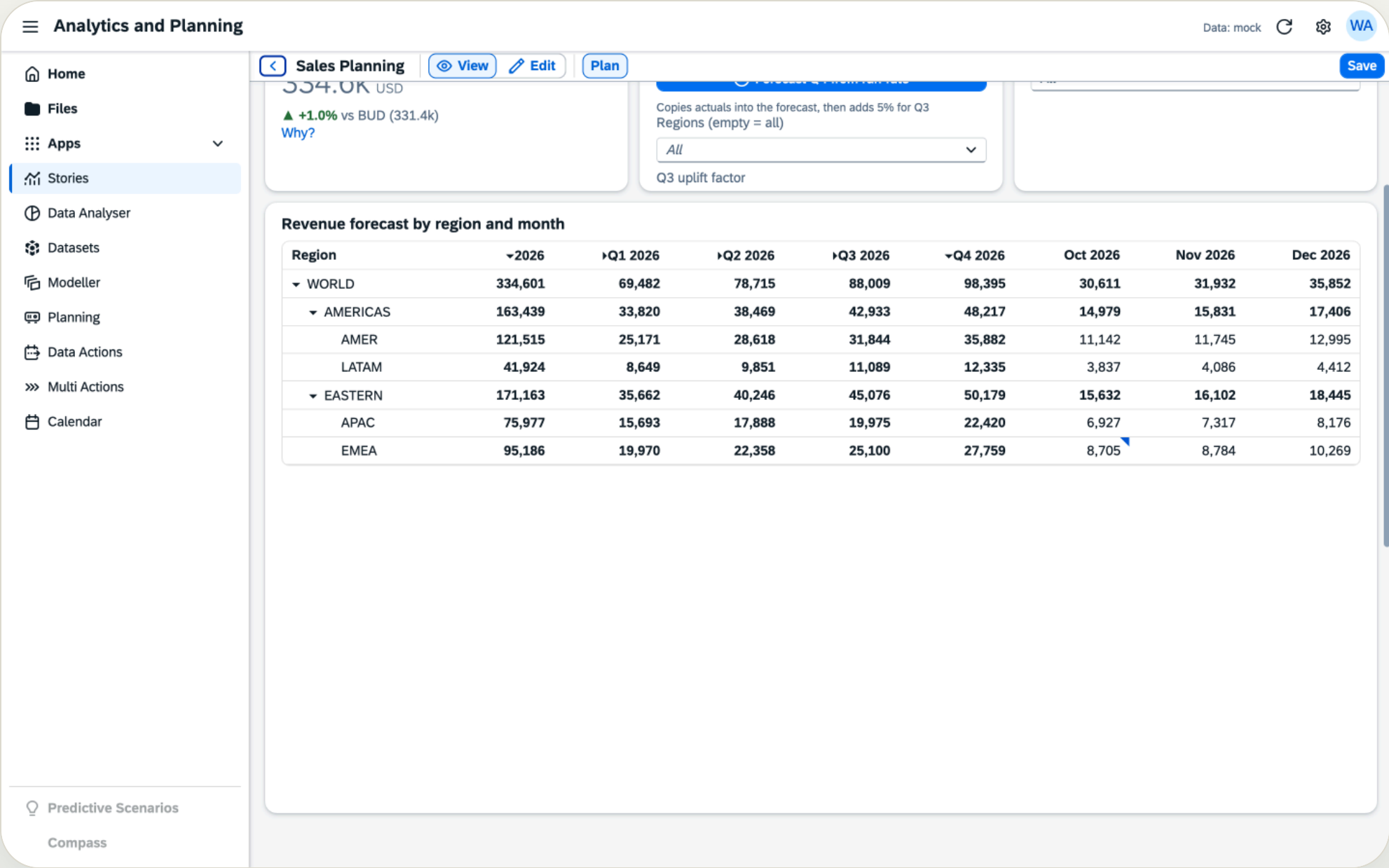
Charts, KPIs and input controls on one page, built in the designer.



SAMPLE DATA

Planning

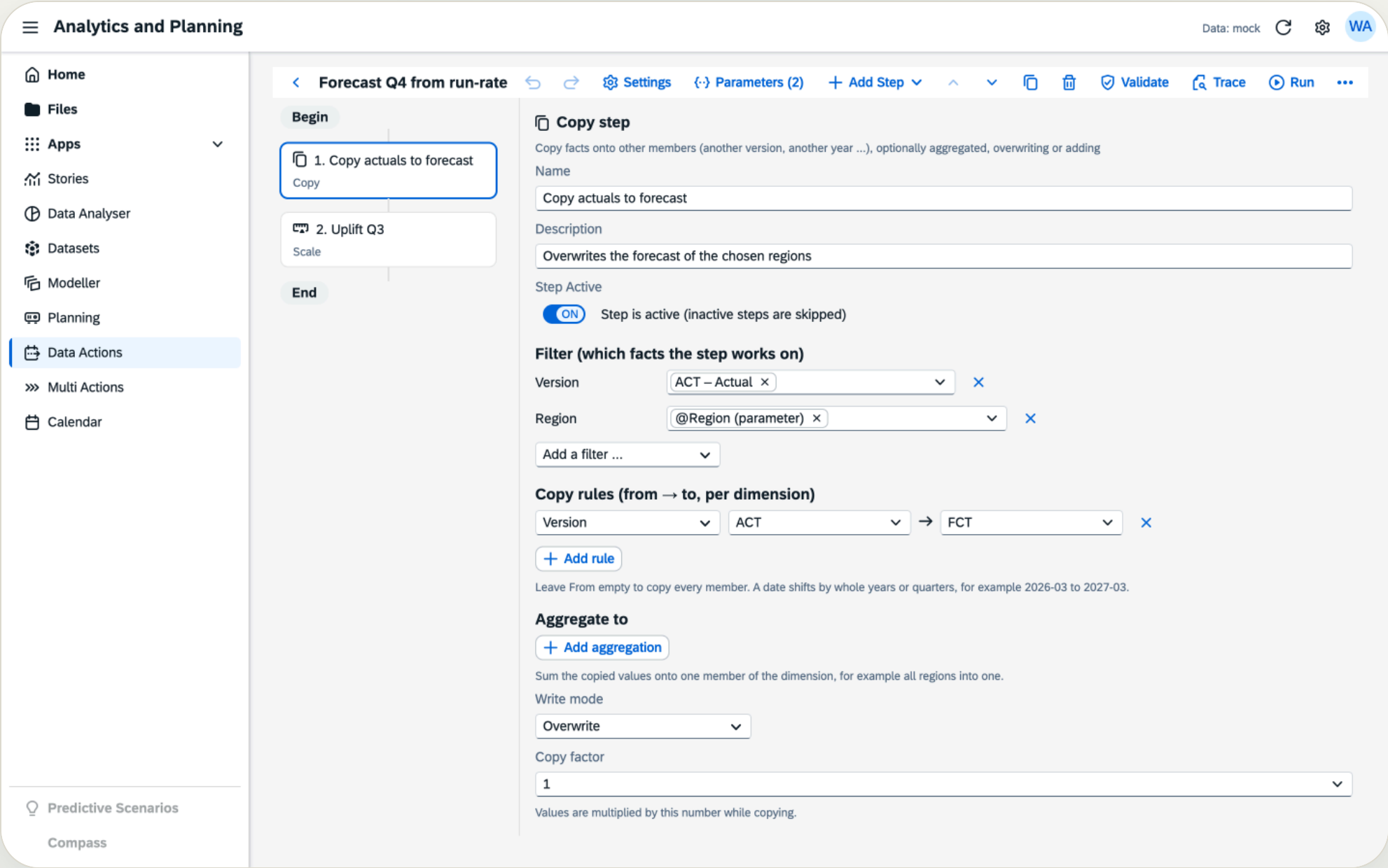
Edit, spread, comment and publish. The orange corner marks a cell with a comment.



SAMPLE DATA

Data actions

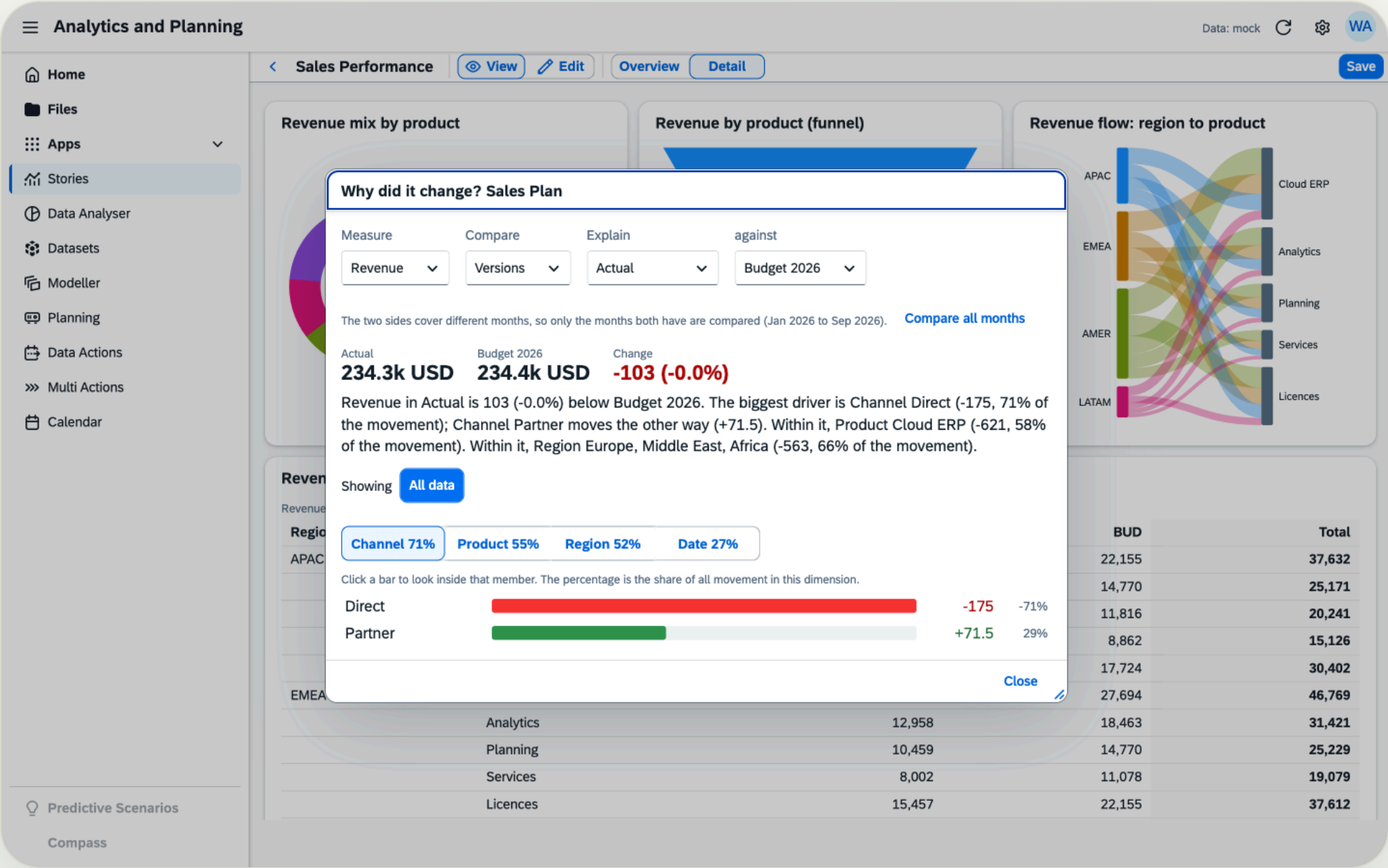
The step flow on the left, the step editor on the right. Validate, trace and run from the toolbar.



SAMPLE DATA

Why did it change?

A headline, the named drivers and ranked dimensions, with bars to drill into.



What was tested

FEATURE	RESULT
Reads, live CDS model, story with filters	Works
Save model, versions, import of 1,092 values	Works
Lock and unlock a version, data kept	Works
Data action with parameters, 329 facts	Works
Multi action, eight steps, 85 seconds	Works
Planning edit, publish, comments, variance	Works
Create private version (server action)	Needs two ABAP lines
API and PaPM steps	As designed

COMPARED WITH SAC

Capability against SAC

CAPABILITY	THIS APP	REAL SAC
Stories, analysis, models	Core set	Complete and mature
Planning with versions	Yes	Plus workflows, approvals
Data actions	Five step kinds	Plus currency, cross-model
Multi actions	Ten step types	Similar, more connectors
Predictive and insights	Forecast, variance	Smart Predict, Just Ask
Scripting, custom widgets	Widget registry	Analytics Designer
Mobile, Excel add-in	No	Yes
Data connections	CDS over OData V4	Many, including BW

COMPARED WITH SAC

Where this app is stronger

Inside the landscape

No extra product, license or tenant. It runs in S/4HANA and the Fiori launchpad, on the user's own authorizations.

Plan data stays in the ERP

No replication. Same transport and backup path, close to the source.

Open to change

Every step, widget and rule can be adapted in days. CDS sources and variance analysis were built for this use.

COMPARED WITH SAC

Where SAC is stronger

Wider scope

Smart Predict, Just Ask, mobile app, Excel add-in and subscriptions are not here. Planning has no workflows or currency conversion.

Scale and speed

Calculations run in the browser. Eight steps on 1,092 rows took 85 seconds; large models will be slower.

Support and governance

We own the code and its upgrades. Sharing rules, row level plan security and content transport are basic.

DECISION

When to use which

Use this app

Data in S/4HANA CDS views, modest planning, no second license, and a fast tailored pilot.

Use real SAC

Enterprise planning with workflows, many users and large volumes, or many source systems.

Use real SAC

Predictive, natural language, mobile and Excel needs, with SAP support and a product roadmap.

NEXT STEPS

Pilot on the G/L model

The feature set is complete on sample data and verified on a real system, except one ABAP fix.

- 1 Apply the ABAP fix and test private versions end to end.
- 2 Decide who sees which story and model, then add authorization.
- 3 Measure with real volumes and decide if actions need a server job.

Decision for Wilmar: run a pilot with one finance team on the G/L model, then decide against SAC from what the users miss.